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CHIRAV DAVE
Pune, India

SUMMARY

AI/ML System Architect specializing in the end-to-end lifecycle of multi-modal models. Skilled in constructing robust fine-tuning pipelines and architecting highly scalable, low-latency deployment infrastructure. Adept at bridging the gap between advanced AI research and enterprise-grade production systems with a focus on reliability and performance.

EDUCATION

MS, CS, Arizona State University | GPA 3.92 *August 2017 - December 2019*

BTech, CS, Maulana Abul Kalam Azad University of Technology | GPA 8.67/10 *July 2012 - July 2016*

TECHNICAL SKILLS

Languages:	Go (Golang), Python, SQL
ML & Inference:	PyTorch, Hugging Face, vLLM, OpenCV
Data & Visualization:	Pandas, Matplotlib
Cloud & Infrastructure:	AWS, Docker, Kubernetes, Helm
Databases:	PostgreSQL, Cassandra, DynamoDB, MySQL, OCI NoSQL

WORK EXPERIENCE

AI/ML Senior Lead, Full-Stack Backend Developer | A10Networks - Incedo | Pune *June 2023*

- Architected an enterprise AI Gateway through general availability, covering Azure IdP integration, user/group controls, provider onboarding, budget and limit configuration, analytics, and distributed logging with data sharding and archival; supported OpenAI, Anthropic, Azure, Cohere, and Gemini.
- Owned end-to-end architecture of an enterprise AI Firewall within a five-engineer team; implemented the management service and distributed logging with partitioning and tracing, supporting paid hardware-integrated deployments across 3–4 customers in Japan.
- Co-defined product vision and feature priorities for the AI Gateway and AI Firewall with engineering leadership, translating customer needs into product capabilities and roadmap decisions.
- Tuned vLLM inference parameters and implemented post-processing to convert LLM outputs into block/allow decisions with reasoning for natural-language custom policies and six OWASP LLM risk categories across prompts and responses; achieved $< 4\%$ false positives and single-digit millisecond added latency under 1,000 concurrent requests with 2,048-character payloads on NVIDIA H100 GPUs.
- Partner with the Cloud Native team to package the AI Gateway & AI Firewall for one-click deployment across Kubernetes environments, simplifying installation and customer onboarding.
- Architected a high-performance data retrieval and request-management system, implementing an enhanced Golang rate-limiting solution that reduced CPU usage by 45%, memory by 38%, and improved response times by 70%, while enabling the targeted filtering and retrieval of 10K attacked IPs from a 65M-entry data warehouse in under 13 seconds.

Technology Stack: PyTorch, HuggingFace, vLLM, GOLANG, FastAPI, DynamoDB, Docker, Helm, Kubernetes

Research Engineer, Lead | DiDi Labs | California *Nov 2021 - April 2023*

- Architected & Implemented a Graph-Based Neural Network to forecast the heading direction of pedestrians within a scene to aid our car's decision-making process.
- Boosted the efficiency of the model from 65% to 72%.
- Optimize and deploy models using techniques such as model compression, pruning, and quantization to improve performance and reduce computational cost.
- Constructed different metrics to measure the accuracy of our algorithms.

Technology Stack: C++, Python, PyTorch

Backend Software Developer | AmazonGO | Washington *Feb 2020 - Sept 2021*

- Designed Java APIs based on SQS to receive requests from clients and utilize vision-based algorithms for predicting the ultimate shopping events.
- Collaborate with data scientists and software engineers to take machine learning models from research and development to production.

Technology Stack: Python, Java, Numpy, Scikit-learn, OpenCV, SQS, Guice

OPEN SOURCE PROJECTS

GPTs from Scratch | Github

- Built a GPT-style model in PyTorch from scratch inspired by Llama, Mistral & DeepSeek for educational purposes.

LLMs for Everyone – Fully Offline | Github

- Developed an open-source chat platform that enables easy offline use of Hugging Face models with integrated back-end and front-end.